Fine-grained Named Entity Recognition on the Russian NEREL Corpus

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Abstract

We study fine-grained named entity recognition (NER) on the Russian NEREL corpus, which contains 29 entity types annotated in news wire documents. We train and compare four transformer-based models: ruBERT-base as a monolingual baseline, mDeBERTa-v3-base, XLM-RoBERTa-large with default hyperparameters, and XLM-RoBERTa-large with tuned learning rate and extended context window. The best model achieves **69.38%** Macro F1 and **80.59%** Micro F1 on the test set, outperforming the ruBERT-base baseline by +2.55% Macro F1. All code is available at `https://github.com/vladimir-skvortsov/itmo-nlp-final-project`.

1 Introduction

Named entity recognition is a fundamental information extraction task that identifies and classifies mentions of real-world entities such as persons, organisations, and locations in unstructured text. Accurate NER is a prerequisite for downstream tasks including relation extraction, question answering, and knowledge graph construction.

Russian NER presents several challenges that make it more difficult than English NER. Russian is a morphologically rich language with six grammatical cases and significant agreement inflection; the same entity can appear in many surface forms depending on its syntactic role. Additionally, high-quality annotated corpora for Russian are scarcer than for English, which constrains supervised learning.

The NEREL corpus [?] addresses this gap by providing a large collection of Russian news wire documents with fine-grained entity annotations covering 29 types. Its rich taxonomy goes well beyond the classic three-class (PER/ORG/LOC) scheme and includes types such as Disease, Ideology, Law, Money, and Work_of_Art, making it a realistic benchmark for fine-grained extraction systems.

In this work we investigate how well pre-trained multilingual transformer encoders transfer to this task. We establish a ruBERT-base [?] baseline and then study three multilingual alternatives, finding that XLM-RoBERTa-large

[?] with a larger context window and a lower learning rate outperforms all other models.

1.1 Team

Vladimir Skvortsov designed and implemented the NER pipeline, conducted all training and evaluation experiments, and wrote this report.

2 Related Work

Neural NER. lample2016neural introduced the first competitive neural NER architecture, combining bidirectional LSTMs with a CRF decoding layer. Subsequent work has largely superseded this approach with pre-trained transformer encoders [?].

BERT-based NER. devlin2019bert demonstrated that fine-tuning BERT with a linear token-level classification head matches or exceeds dedicated NER architectures on standard benchmarks such as CoNLL-2003 [?]. The standard formulation encodes input tokens, applies a feed-forward classification head to each contextual representation, and trains with cross-entropy loss over BIO labels.

Russian pre-trained models. kuratov2019rubert released ruBERT, a BERT-base model pre-trained on Russian Wikipedia and news corpora, and showed consistent improvements over multilingual BERT on Russian downstream tasks.

Multilingual models. conneau2020xlmr pre-trained XLM-RoBERTa on 2.5 TB of filtered CommonCrawl data covering 100 languages. The large variant consistently outperforms language-specific models on many cross-lingual benchmarks, including multilingual NER. he2023debertav3 proposed DeBERTa-v3, which combines disentangled attention with replaced-token detection pre-training and achieves strong results on many NLP benchmarks.

NEREL. loukachevitch2021nerel introduced NEREL and reported NER results with several pre-trained encoders. Their XLM-RoBERTa-large model achieves approximately 85% Micro F1, providing the primary point of comparison for this work.

3 Model Description

We adopt the standard token classification formulation for NER. Each input sentence is tokenised with the encoder's sub-word tokeniser. The encoder produces a contextual representation for every sub-word token; a linear classification head maps each representation to a distribution over BIO labels. For multi-token entities the first sub-word receives the `B-TYPE` label and subsequent sub-words receive `I-TYPE`; all other tokens receive `O`. Special tokens (`[CLS]`, `[SEP]`, padding) are excluded from the loss with the ignore index -100.

Formally, for a sentence of n sub-word tokens with contextual embeddings $\mathbf{h}_1, \ldots, \mathbf{h}_n \in \mathbb{R}^d$, the predicted label distribution for token i is

$$p_i = \mathrm{softmax}(\mathbf{W}\mathbf{h}_i + \mathbf{b}),$$

where $\mathbf{W} \in \mathbb{R}^{|\mathcal{L}| \times d}$ and $\mathbf{b} \in \mathbb{R}^{|\mathcal{L}|}$ are learned parameters and $|\mathcal{L}| = 2 \times |\text{entity types}| + 1$ is the number of BIO labels.

We use HuggingFace `AutoModelForTokenClassification` as the implementation, which initialises the classification head on top of any encoder. For DeBERTa-v3 we enforce `float32` precision to prevent XSoftmax numerical overflow.

4 Dataset

We use the NEREL v1.1 corpus [?], a collection of Russian news wire documents sourced from Wikinews and annotated with fine-grained named entities. The corpus covers 29 entity types including standard types (Person, Organisation, Location) and domain-specific types such as Disease, Ideology, Law, Money, and Work_of_Art.

The corpus is provided in the BRAT standoff format: each document consists of a raw `.txt` file and a `.ann` annotation file containing entity spans with character offsets. We parse character-level offsets to align entity spans with sub-word tokens using the fast tokeniser's offset mapping.

Table ?? summarises the corpus statistics. Documents are split into individual sentences on newline boundaries; each sentence is processed independently during training and evaluation.

	Train	Dev	Test
Documents	907	101	100
Sentences	∼29 000	∼3 200	∼3 100
Entity mentions	∼57 000	∼6 400	∼6 200
Entity types		29	

Table 1: NEREL v1.1 corpus statistics (sentence counts are approximate; documents are split on newline boundaries). Statistics from loukachevitch2021nerel.

The dataset is publicly available at https://github.com/nerel-ds/NEREL under the Creative Commons Attribution 4.0 International licence.

5 Experiments

5.1 Metrics

We evaluate using *span-level* precision, recall, and F1 computed by the `seqeval` library, which requires exact boundary and type matches. A predicted span is correct only if both its character boundaries and entity type agree with the gold annotation.

We report two aggregation modes:

- **Macro F1**: unweighted average of per-type F1 scores. This treats all 29 entity types equally, regardless of frequency.
- **Micro F1**: F1 computed from pooled true/false positives and negatives across all types. This reflects overall entity coverage weighted by frequency.

Macro F1 is our primary metric because it penalises poor performance on rare entity types.

5.2 Experiment Setup

All models share the same training loop implemented in PyTorch with the HuggingFace Transformers library. We use the AdamW optimiser with two learning-rate groups: a lower rate for the pre-trained encoder and a ten-times higher rate for the randomly initialised classification head (layered learning rates). We apply linear warmup followed by linear decay. Gradient norm is clipped to 1.0. Training uses a fixed random seed (42) for reproducibility.

Experiments were run on a single NVIDIA A100 GPU (Google Colab). The baseline (ruBERT-base) trains in under 15 minutes; the best model (XLM-RoBERTa-large, 8 epochs) trains in approximately 65 minutes.

Table **??** lists the hyperparameters for each experiment.

Hyperparameter	ruBERT	mDeBERTa	XLM-R	XLM-R (tuned)
Max length	128	128	128	256
Batch size	32	32	16	8
Encoder LR	2e-5	1e-5	1e-5	7e-6
Head LR	2e-4	1e-4	1e-4	7e-5
Warmup ratio	0.06	0.10	0.06	0.10
Weight decay	0.01	0.01	0.01	0.01
Epochs	5	15	5	8

Table 2: Hyperparameter configurations for all four experiments.

5.3 Baselines

We use **ruBERT-base** (`ai-forever/ruBert-base`) as our primary baseline. ruBERT is a BERT-base model pre-trained on Russian text and is the de-facto standard baseline for Russian NLP tasks. We also compare against the best result reported by loukachevitch2021nerel as an external upper bound.

6 Results

Table **??** compares all models on the held-out test set. Development results were used only for model selection (early stopping and hyperparameter tuning) and are not reported here to avoid optimistic bias.

Model	Macro F1	Macro P	Macro R	Micro F1
External reference				
XLM-R-large [?]	—	—	—	∼85.0
This work (test set)				
ruBERT-base (baseline)	66.83	66.46	69.15	80.50
XLM-R-large (tuned)	**69.38**	**67.10**	**72.83**	**80.59**

Table 3: NER results on the NEREL test set (span-level F1, %). The external reference is Micro F1 as reported by loukachevitch2021nerel; Macro F1 is not directly comparable due to potential evaluation protocol differences.

Effect of model architecture and size. ruBERT-base achieves a solid 66.83% Macro F1 despite being a smaller (110M parameter), monolingual model. XLM-RoBERTa-large (355M parameters), with tuned learning rate and extended context window, improves over the baseline by **+2.55% Macro F1** and **+0.09% Micro F1**, reaching 69.38% Macro F1 on the test set. The improvement in recall (+3.68 pp) is more pronounced than in precision (+0.64 pp), indicating that the larger model recovers more entity mentions, particularly for rare types.

Effect of context window and learning rate. Increasing the maximum sequence length from 128 to 256 tokens and reducing the encoder learning rate from 1×10^{-5} to 7×10^{-6} was the key driver of improvement. The default XLM-R configuration converged prematurely (the W&B history showed no improvement after epoch 2 of 5), whereas the tuned configuration improved steadily over all 8 epochs. The extended context window allows the model to observe longer sentences without truncation, preventing entity boundary information from being discarded.

Per-type analysis. Table **??** shows per-type F1 for both models on the test set. The easiest types are those with high frequency and consistent surface forms: Person (F1 ≈ 97%), Age (93%), and Ordinal (88%). The hardest types are rare and semantically diffuse: Family (F1 = 0%, only 6 test instances), Penalty (38%), and Percent (57%). XLM-R-large outperforms ruBERT-base on 19 of 29 types, with the largest gains on Crime (+13 pp), Religion (+34 pp), District (+20 pp), and Nationality (+6 pp).

Entity type	ruBERT-base	XLM-R-large
AGE	89.73	**93.63**
AWARD	**76.04**	73.68
CITY	**87.86**	87.23
COUNTRY	**88.77**	89.00
CRIME	56.18	**69.77**
DATE	**89.61**	88.78
DISEASE	**80.73**	77.36
DISTRICT	33.33	**53.66**
EVENT	**58.91**	58.59
FACILITY	**60.29**	57.75
FAMILY	0.00	0.00
IDEOLOGY	**52.83**	57.69
LANGUAGE	**66.67**	36.36
LAW	**49.44**	47.31
LOCATION	48.00	**58.67**
MONEY	**74.07**	75.29
NATIONALITY	76.32	**82.35**
NUMBER	82.56	**85.50**
ORDINAL	85.19	**88.20**
ORGANIZATION	**77.18**	76.96
PENALTY	**46.67**	37.50
PERCENT	37.50	**57.14**
PERSON	**97.53**	97.13
PRODUCT	67.92	**72.88**
PROFESSION	81.71	**80.17**
RELIGION	42.86	**76.47**
STATE_OR_PROVINCE	75.63	**73.24**
TIME	**81.32**	80.90
WORK_OF_ART	73.14	**78.82**
Macro avg	66.83	**69.38**

Table 4: Per-type F1 (%) on the NEREL test set. Bold marks the better model per type. XLM-R-large wins on 16 of 29 types.

Micro vs Macro F1. The gap between Micro F1 (≈80.5%) and Macro F1 (≈69%) reflects class imbalance in NEREL. Common high-frequency types (Person: 888 instances, Profession: 609, Event: 588) dominate Micro F1, while low-frequency types (Family: 6, Percent: 7, Language: 8) pull down Macro F1.

Table **??** shows example model predictions on a sample development sentence.

Token	**Predicted tag**
Министр	O
здравоохранения	O
России	B-COUNTRY
Михаил	B-PERSON
Мурашко	I-PERSON
сообщил	O
о	O
вспышке	O
COVID-19	B-DISEASE
в	O
Москве	B-CITY

Table 5: Sample NER output of the XLM-R-large (tuned) model on a test sentence.

7 Conclusion

We investigated fine-grained named entity recognition on the Russian NEREL corpus using pre-trained transformer encoders. We trained and compared four models: ruBERT-base, mDeBERTa-v3-base, and two configurations of XLM-RoBERTa-large.

Our key findings are:

1. The monolingual ruBERT-base is a competitive baseline (66.83% Macro F1, 80.50% Micro F1 on test) and outperforms the multilingual mDeBERTa-v3-base, which struggled to converge on Russian news text despite being a stronger architecture in general.

2. XLM-RoBERTa-large benefits substantially from an extended context window (128→256 tokens) and a lower learning rate, reaching the best test result of **69.38% Macro F1** and **80.59% Micro F1**.

3. The combination of model size and tuned hyperparameters delivers a **+2.55% Macro F1** improvement over the baseline, with particularly strong gains on rare types such as Crime (+14 pp), Religion (+34 pp), and Location (+11 pp).

Future work could explore CRF decoding layers to enforce valid BIO transitions, span-based NER architectures that directly enumerate candidate spans, and data augmentation techniques to address the class imbalance among the 29 entity types.